

Tabular Planning: Value Iteration

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CS 1840 Fall 2026

Announcements

HW1 is out!

HW0 is due Sept 17 (**this Thursday**) , 11:59 pm ET

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: Fixed-point solution

Find the fixed point solution of $x^\star = f(x^\star)$, $x \in \mathbb{R}$



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Start with some x_0 , set $x_{t+1} \Leftarrow f(x_t)$

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Suppose f is contraction, i.e., $\forall x, x', |f(x') - f(x)| \leq \gamma |x' - x|, \gamma \in [0, 1)$,
then $x_t \rightarrow x^\star$

Finishing Up From Last Lecture



1. Exact Policy Evaluation



2. Approximate Policy Evaluation via an Iterative Algorithm

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0,1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

$$\|V\|_\infty = \max_{s \in \mathcal{S}} |V(s)|$$

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$$\forall s, \left| \underset{\triangle}{V^{t+1}}(s) - \underset{\triangle}{V^\pi(s)} \right|$$

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$$\begin{aligned} & \forall s, \left| V^{t+1}(s) - V^\pi(s) \right| \\ &= \left| \underbrace{r(s, \pi(s))}_{\text{By Alg}} + \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s')}_{\Delta} - \left(\underbrace{r(s, \pi(s))}_{\Delta} + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \underbrace{V^\pi(s')}_{\Delta} \right) \right| \end{aligned}$$

$\nwarrow \text{By Alg}$
 $\nwarrow \gamma \mathbb{E}$

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$$\forall s, \left| V^{t+1}(s) - V^\pi(s) \right| \quad r(s, \pi(s)) \quad \pi(s) \rightarrow a$$

$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

$$= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right|$$

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$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|$$



$$\begin{aligned} & \left| \mathbb{E}_{f \sim P} f(x) - \mathbb{E}_{g \sim P} g(x) \right| \\ & \leq \mathbb{E}_{x \sim P} |f(x) - g(x)| \end{aligned}$$

Convergence of Iterative PE

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$$= \gamma \left| \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right|$$

$$\leq \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|}_{\forall s} \quad \text{Avg} \leq \max \quad \mathbb{E}_{x \sim p} |f(x)| \leq \max_x |f(x)|$$

$$\leq \gamma \left\| V^t - V^\pi \right\|_\infty$$

Convergence of Iterative PE

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$$\forall s, \left| \underline{V^{t+1}(s)} - \underline{V^\pi(s)} \right|$$

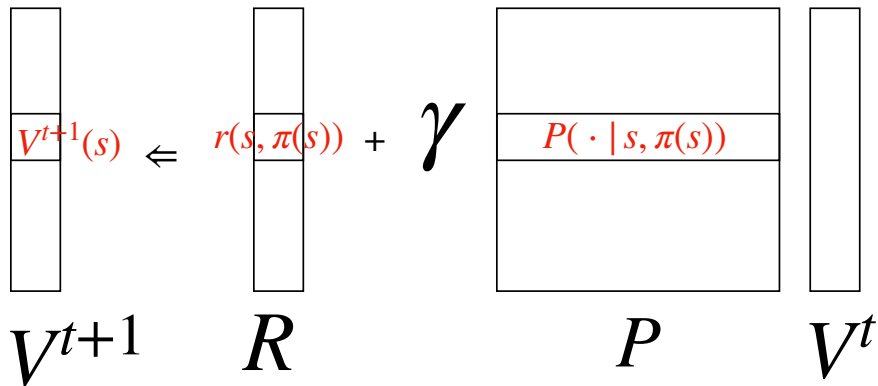
$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} V^\pi(s') \right) \right|$$

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$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi(s))} \left| V^t(s') - V^\pi(s') \right|$$

$$\leq \gamma \left\| \underline{V^t - V^\pi} \right\|_\infty \Rightarrow \left\| \underline{V^{t+1} - V^\pi} \right\|_\infty \leq \gamma \left\| \underline{V^t - V^\pi} \right\|_\infty \leq \dots \leq \gamma^{t+1} \left\| V^0 - V^\pi \right\|_\infty$$

Summary so far:



Convergence:

$$\| V^{t+1} - V^\pi \|_\infty \leq \gamma \| V^t - V^\pi \|_\infty \leq \gamma^{t+1} \| V^0 - V^\pi \|_\infty$$

Recap: Optimal Policy $\pi^\star$

For infinite horizon discounted MDP, there always exists a deterministic policy

$$\pi^\star : S \mapsto A, \text{ s.t., } \underline{V^{\pi^\star}}(s) \geq V^\pi(s), \forall s, \pi$$

[Puterman 94 chapter 6, also see theorem 1.7 in the RL monograph—no need to understand the proof]

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We often denote $V^\star, Q^\star$ in short for $V^{\pi^\star}, Q^{\pi^\star}$



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$V^\star(s)$: the maximum value we could possibly achieve

Question for Today:

Given an MDP $\mathcal{M} = (S, A, P, r, \gamma)$,
How to find Q^* and π^* (approximately)

Motivation for Finding the Optimal Policy

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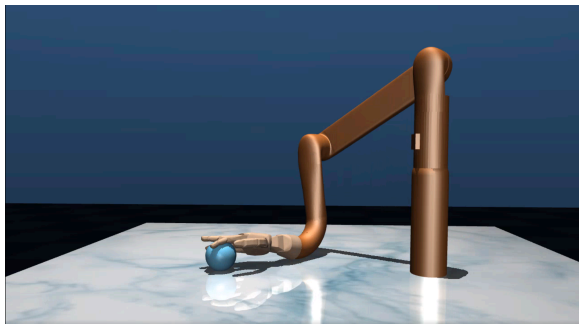
Find the strategy w/ the highest
prob of winning
(i.e., a policy that maps the board
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Motivation for Finding the Optimal Policy



Find the strategy w/ the highest
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Find the strategy (i.e., a mapping from
robot & ball configuration to torques)
that picks the ball and moves it to a
goal position ASAP

Outline:

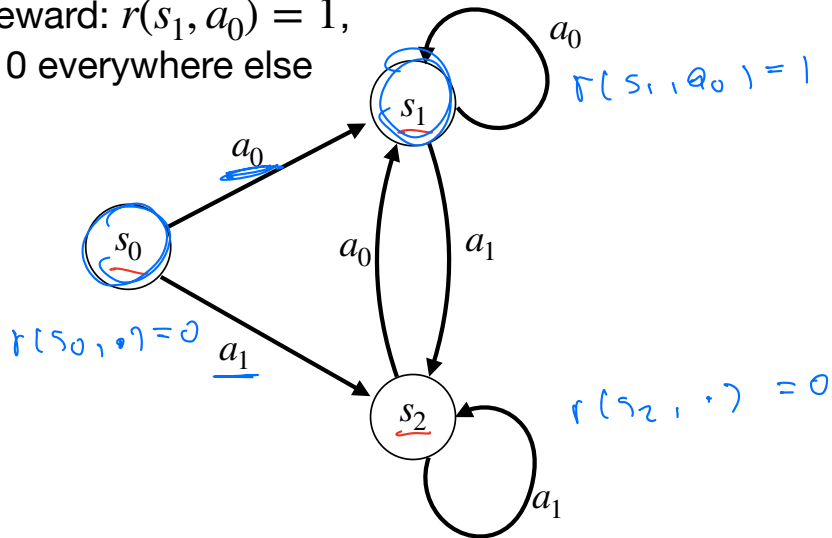
1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

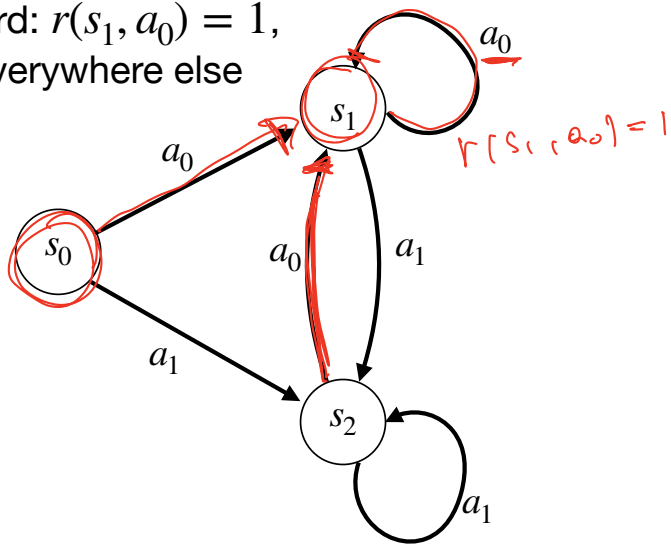
Reward: $r(s_1, a_0) = 1$,
0 everywhere else



Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

Reward: $r(s_1, a_0) = 1$,
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Let's say $\gamma \in (0,1)$
What's the optimal policy?

$$\pi^\star(s_0) = a_0$$

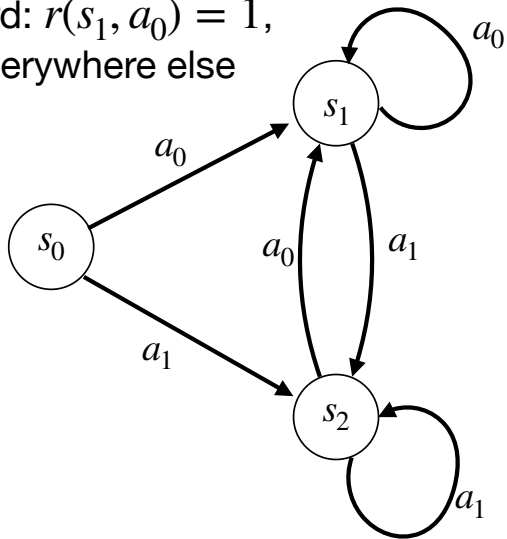
$$\pi^\star(s_1) = a_1$$

$$\pi^\star(s_2) = a_1$$

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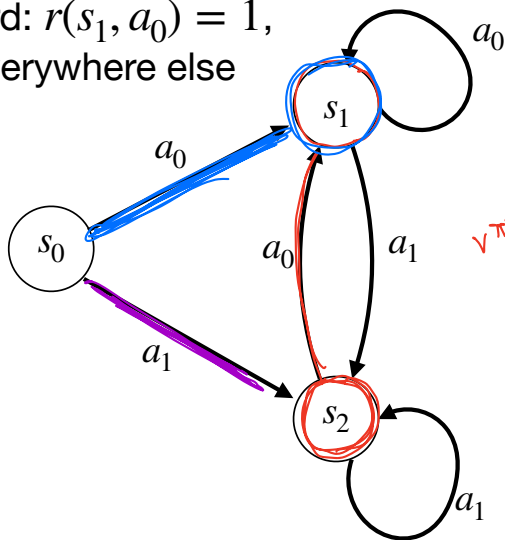
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Example of Optimal Policy $\pi^\star$

Consider the following **deterministic** MDP w/ 3 states & 2 actions

Reward: $r(s_1, a_0) = 1$,
0 everywhere else



If we were told $V^\star(s_1)$ & $V^\star(s_2)$,
how to compute $V^\star(s_0)$

$$V^\pi(s) = \mathbb{E}_{s \sim p} \left[\sum_{n=0}^{\infty} \gamma^n r_n \right]$$

$$V^{\pi^\star}(s_1) = 1 + \gamma \cdot 1 + \gamma^2 + \dots$$

$$= \frac{1}{1-\gamma}$$

$$V^{\pi^\star}(s_2) = 0 + \gamma \cdot 1 + \gamma^2 + \dots$$

$$= \gamma(1 + \gamma + \gamma^2 + \dots)$$

$$= \frac{\gamma}{1-\gamma}$$

$$V^{\pi^\star}(s_0) = \max_a \left\{ r(s_0, a_0) + \gamma V^{\pi^\star}(s_1), \right. \\ \left. r(s_0, a_1) + \gamma V^{\pi^\star}(s_2) \right\}$$

$$V^* = V^{\pi^*}$$

Bellman Optimality

Value function of
the optimal π^*

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right], \forall s$$

Bellman Optimality

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$$V^*(s) = \left[r(s, \pi^*(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \pi^*(s))} V^*(s') \right] \quad (\text{By BE of } \pi^*)$$

$\pi^*(s) \rightarrow a$

Bellman Optimality

Bellman Optimality

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$$V^*(s) = \left[r(s, \underline{\pi^*(s)}) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, \underline{\pi^*(s)})} V^*(s') \right] \quad (\text{By BE of } \pi^*)$$

$$\leq \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right]$$

Bellman Optimality

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$> V^*(s)$

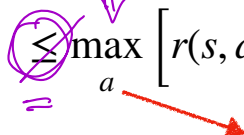
If we took this $\arg \max_a$ at s , then follow π^* , we **would have higher value**

Bellman Optimality

Bellman Optimality

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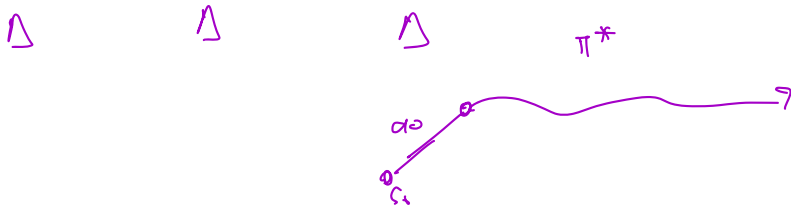
Contradicts to the fact that $V^*(s)$ is the maximum value at s one could possibly achieve

Bellman Optimality

Bellman Optimality

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

Recall that $Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} V^\star(s')$



Bellman Optimality

Bellman Optimality

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s') \right]$$

Recall that $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s')$

This implies that $\arg \max_a Q^*(s, a)$ is an optimal policy

$$V^*(s) = \max_a [Q^*(s, a)]$$

$$\pi = \arg \max_a Q^*(s, a)$$



Bellman Optimality

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An optimal policy should pick
this action at s

Outline:



1: Bellman Optimality

2: An Iterative Algorithm: Value Iteration

Define the Bellman optimality for $Q^\star$

We now know:

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

(Note: The original image contains handwritten purple annotations: a triangle under the max operator, and underlines under the terms $r(s, a)$, γ , and $\mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s')$.)

What's the Q version?

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} [V^\pi(s')]$$

(Note: The original image contains handwritten purple annotations: a circled π in the superscript of V^π , and a subscript $s' \sim p$ under the expectation operator.)

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What's the Q version?

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

Define the Bellman optimality for $Q^\star$

We now know:

$$V^\star(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^\star(s') \right]$$

$f(f(f(Q^\star)))$

What's the Q version?

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

To estimate $Q^\star$, we will use the fix-point iterative approach again

Define the Bellman operator

Given a function $Q : S \times A \mapsto \mathbb{R}$,

$$\mathcal{T}Q : S \times A \mapsto \mathbb{R},$$

$$(\mathcal{T}Q)(s, a) := \underline{r(s, a)} + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in A} Q(s', a') \right], \quad \forall s, a \in S \times A$$

$$\begin{array}{l} s \mapsto \\ a \mapsto \end{array} \boxed{\mathcal{T}Q} \Rightarrow r(s|a) + \gamma \mathbb{E}_{s' \sim P} \max_{a' \in A} Q(s'|a')$$

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We can express $Q \in \mathbb{R}^{|S||A|}$, so $\mathcal{T}Q \in \mathbb{R}^{|S||A|}$

A handwritten diagram in purple ink illustrating the Bellman operator's action on a Q-value vector. On the left, a vertical vector bracket is labeled $Q(s, a)$ and $\in \mathbb{R}^{|S||A|}$. An arrow points from this vector to a similar vector on the right. The right-hand vector is labeled with the expression $r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in A} Q(s', a')$ and also $\in \mathbb{R}^{|S||A|}$.

Define the Bellman operator

Given a function $Q : S \times A \mapsto \mathbb{R}$,

$$\mathcal{T}Q : S \times A \mapsto \mathbb{R},$$

$$\begin{aligned} & \max_x (f(x) + g(x)) \\ \Rightarrow & \max_x f(x) + \max_x g(x) \end{aligned}$$

$$(\mathcal{T}Q)(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in A} Q(s', a'), \forall s, a \in S \times A$$

We can express $Q \in \mathbb{R}^{|S||A|}$, so $\mathcal{T}Q \in \mathbb{R}^{|S||A|}$

i.e., think about $\mathcal{T}$ as a (non-linear) mapping that maps from $\mathbb{R}^{|S||A|}$ to $\mathbb{R}^{|S||A|}$

High Level idea for Algorithm Design

Fix-point iteration again!

High Level idea for Algorithm Design

Fix-point iteration again!

Recall Bellman Optimality for $Q^\star$:

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$


High Level idea for Algorithm Design

Fix-point iteration again!

Recall Bellman Optimality for $Q^\star$:

$$Q^\star(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} Q^\star(s', a')$$

We have $Q^\star = \mathcal{T} Q^\star$,
i.e., $Q^\star$ is a fix-point solution of $Q = \mathcal{T} Q$

Value Iteration Algorithm:

1. Initialization: $Q^0 \in [0, \frac{1}{1-\gamma}]$

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

$$r \in [0, 1]$$

$$\gamma \in [0, 1)$$

$$1 + \gamma + \gamma^2 + \dots$$

$$= \frac{1}{1-\gamma}$$

Value Iteration Algorithm:

1. Initialization: Q^0

2. Iterate until convergence: $Q^{t+1} \Leftarrow \mathcal{T} Q^t$

We hope $Q^t \rightarrow Q^\star$, as $t \rightarrow \infty$

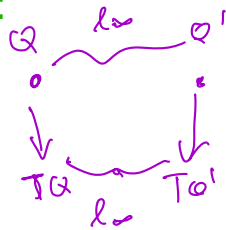
Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_{\infty} \leq \gamma \|Q - Q'\|_{\infty}$$

Proof:

$$l_{\infty}: \|x\|_{\infty} = \max_i |x_i|$$



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Proof:

$\forall s, a$

$$|(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| = \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right|$$

Handwritten notes: "by definition" with arrows pointing to the expectation terms.

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$$\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right|$$

$$| \mathbb{E}_{x \sim p} f(x) - \mathbb{E}_{x \sim p} g(x) |$$

$$\leq \mathbb{E}_{x \sim p} |f(x) - g(x)|$$

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$\forall s, a$

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \leq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} |Q(s', a') - Q'(s', a')| \end{aligned}$$

$$\begin{aligned} & \max_x |f(x) - g(x)| \\ & \leq \max_x |f(x) - g(x)| \end{aligned}$$

Convergence of Value Iteration:

Lemma [contraction]: Given any Q, Q' , we have:

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty$$

Proof:

$$\begin{aligned} |(\mathcal{T}Q)(s, a) - (\mathcal{T}Q')(s, a)| &= \left| r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q(s', a') - \left(r(s, a) + \gamma \mathbb{E}_{s' \sim P(s, a)} \max_{a'} Q'(s', a') \right) \right| \\ &\leq \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}}_{\text{Avg}} \left| \left(\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right) \right| \leq \underbrace{\gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a'} |Q(s', a') - Q'(s', a')|}_{\text{Avg} \leq \max} \\ &\leq \gamma \underbrace{\max_{s'} \max_{a'}}_{\text{Avg} \leq \max} |Q(s', a') - Q'(s', a')| \end{aligned}$$

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$$\begin{aligned} \forall s, a \quad |TQ(s, a) - TQ'(s, a)| &\leq \gamma \|Q - Q'\|_\infty \\ \max_s \max_a |TQ(s, a) - TQ'(s, a)| &\leq \gamma \|Q - Q'\|_\infty \\ \|TQ - TQ'\|_\infty &\leq \gamma \|Q - Q'\|_\infty \end{aligned}$$

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Lemma [Convergence]: Given Q^0 , we have:

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$$\dots \leq \gamma^{t+1} \|\widehat{Q}^0 - Q^\star\|_\infty$$

Summary so far:

VI (a fix point iteration alg):

$$Q^{t+1} \Leftarrow \mathcal{T} Q^t$$

VI convergence (via contraction)

$$\text{i.e., } \|Q^t - Q^\star\|_\infty \leq \gamma^t \|Q^0 - Q^\star\|_\infty$$

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What about the policy? Ultimately, we do want $\pi^\star$...



Summary

Attendance

<https://bit.ly/46pCUdN?r=qr>



Feedback

<https://bit.ly/4gYn2nC>

